BrickPlan: Visual Assembly and Repair Beyond a Single Build Order Supplementary Material

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A. Artifact Scope and Reproducibility

The manuscript describes an evidence snapshot dated 21 October 2026. Three banks coexist: the 3,292-question whiteboard static bank, the 1,872-episode dynamic mechanism bank, and the 96-case community module suite. Their counts are not independent model populations, and their experimental results are not interchangeable. The new route study runs on the community module suite, not on the historical six-part car. The complete module suite has no formal learned-agent evaluation or independent human assembly experiment in this snapshot.

The artifact records original model files, source metadata, certificates, cases, reference manuals, intervention witnesses, and order-regression checks. Each instruction contains the physical part and the resulting scene. Image manifests bind the current benchmark renders to their source states. Paper figure provenance separately records source hashes, cameras, native render dimensions, and exported figure hashes. The deterministic study records the four input files for each object and their SHA-256 digests. Re-execution does not require a model API.

The public object gallery is an author-facing inspection tool: it shows fault names and correction witnesses. It is not a valid blind agent input. Evaluation must use the observation adapter, which exposes the permitted images and action handles while retaining certificates, intervention labels, event timing, and recovery witnesses privately. Publishing such annotations supports reproducibility but also makes these public examples unsuitable as an undisclosed held-out test set.

B. Exact Observation and Action Roles

Table S1 distinguishes the images that define a task from those that show its current state. In a static question, the acceptance target is an intermediate stage. A full-completion reference is provided for context and must not silently replace that stage target. In a dynamic episode, the acceptance target is the completed module and remains fixed even when a hidden intervention occurs.

Views and handles. The module interface provides isometric, rear, and top views at 900×660 pixels. Part and candidate handles are randomized by seed and do not expose target labels. The manual stays accessible throughout the episode. Current-state images do not include the paper's error boxes or corrective arrows. Explicit request and response logs distinguish inspection from manipulation and retain the actual view delivered to the agent.

One physical action per turn. An attach action chooses one rack piece and one catalog candidate. A detach action removes one installed piece. The executor checks the resulting configuration and a straight insertion or extraction path. An invalid physical attempt consumes a manipulation attempt, reports a legality code, and leaves the world unchanged. Inspection does not advance construction or trigger an event. Finish requests termination; success additionally requires the exact permitted geometric endpoint. The manipulation budget is $4N + 12$ for N normal module pieces. The outer model runner has a separately configured decision cap so repeated inspection cannot run indefinitely. Both limits must be reported in a model comparison.

Finite action candidates. The action catalog is part of the task definition. This is not unrestricted six-degree-of-freedom pose prediction. Candidate previews and legality codes reduce the planning burden and can leak useful physical information, even though neither states the correction. They must be held constant across compared agents; an ablation can quantify their contribution. A symbolic state or expected next-state overlay is privileged information and should be reported as a separate condition.

Indistinguishable parts. Terminal comparison uses physical part geometry, color, and pose rather than instance identity. Supported rotational equivalences are limited to plain rectangular parts with the appropriate geometric symmetry. Decorated, directional, and asymmetric pieces are not assumed invariant to arbitrary rotations. Identical-part

Table S1. **Image roles in the community interface.** A reference image is not a report of current execution. These roles are explicit in the adapter rather than inferred from file order.

Role	Content	Interpretation
Scene overview	Full original object, initial and completed views	Establishes the recognizable object and the supplied context.
Module start	Cropped scene before any task manipulation	The initial movable-part state; the context is already assembled.
Full-completion reference	Completed remaining module	Explains the whole module; may differ from a static question's acceptance target.
Acceptance target	Required final state for this episode	Stage target for static repair; completed module for dynamic completion.
Instruction part / after	Isolated part and scene after its reference placement	One executable demonstration; the numbered order is not compulsory.
Inventory card	Shape and color of an available piece	Identifies an action handle without declaring its correct placement.
Before last action	Actual scene before the agent's last action	Temporal evidence; explicitly absent at the first decision.
Current scene	Actual state after execution and any intervention	The authoritative observation of the world now.
Candidate preview	Hypothetical placement of a selected piece	A planning aid; not executed, not a correctness certificate.

exchanges are tested separately from sequence exchanges: changing an identifier alone must not create an apparent alternative physical trajectory.

C. Intervention Selection and Causal Comparisons

Deterministic adaptive selection. Each community fault condition names one of three mechanisms and an installed-part threshold. After an accepted manipulation reaches the threshold, the selector inspects the actual state. For a missing part or color replacement, it considers older installed normal pieces, retaining the newest piece to avoid merely undoing the last action. For an extra layer, it searches compatible exposed positions for the spare plate. The control and all three fault conditions share the same inventory and candidate catalog.

An event is admitted only if it changes the state through legal primitives and has a feasible inverse returning to the pre-event state. A replacement uses a same-shape alternative-color spare: the environment removes the original piece and attaches the spare between observations. These two environment primitives do not allow the agent to submit two manipulations in one turn. If no suitable event is available, selection is deferred. At most one event is applied in the current module suite. The private log records the realized location and its inverse.

What feasibility establishes. A legal inverse is a recovery witness, not a proof that it is the shortest recovery or that the error is visually obvious. Every selected normal route in the protocol study has a complete replay, so restoring its pre-event state permits the remaining route to finish. This guarantee does not extend to an arbitrary agent state that was already incorrect before the intervention. Likewise, a reference event's RGB check does not certify visibility at every location an adaptive selector can reach. New trajectories require their own observation-level audit.

Matched-prefix controls. For a causal recovery comparison, execute a normal prefix once, save the environment and conversation, and branch it into control and fault continuations. The two branches have identical history before the event. Keep the target, action vocabulary, inventory, renderer, budgets, and model settings fixed. Report both completion and whether the fault actually occurred. A run that finishes correctly without encountering its scheduled fault is a completion, not a successful repair trial.

A conditional recovery rate excludes untriggered episodes from its denominator, so it must be accompanied by exposure coverage and the number of stalled prefixes. Even with the same input seed, two fresh API runs are not necessarily matched-prefix controls: their actions may diverge before the event. Adaptive location selection also creates trajectory-dependent fault difficulty. Stratify by the realized edit and report the reference-location subset separately when appropriate.

D. Community Objects and Route Diagnostics

Table S2 lists all twelve objects. Original instance counts include the supplied surrounding scene, figures, and accessories. The module count measures the actual task; it does

not imply reconstructing the entire source object. The full scenes contain 2,135 source instances; the bounded modules contain 99 normal placements. Larger source models therefore need not be harder tasks.

Selection of the alternative routes. Each reference route is the verified manual. An alternative is obtained by testing swaps and retaining a physically legal route with all three feasible adaptive faults. When available, the search favors a route whose geometry at the threshold differs from the reference checkpoint. The study therefore uses selected regression examples, not a random sample of all topological orders or an estimate of agent route choice. Table S3 provides the precise reference-position indices.

Exact-prefix opportunity. The comparison inspects the otherwise normal physical trajectory for equality with the reference checkpoint. This avoids confounding trigger eligibility with a different repair policy. It yields 36/36 eligible reference-route fault trials and 9/36 alternative-route trials. The actual-state selector applies an event in all 36 trials in each group. A separate comparator requiring every intermediate state to match the manual accepts 12/12 reference routes and 0/12 alternatives. The terminal comparator accepts all 24 normal routes. These numbers establish counterexamples to an overly restrictive contract, not a learned-agent performance improvement.

The blind policy follows its normal route after any change. It completes 24/24 controls and 0/72 fault trials. The privileged policy uses the recorded inverse immediately after the event and then continues; it completes 96/96 trials. No model API is involved. A privileged repair trace tests solvability and logging, not visual detection, reasoning, or human-level assembly skill.

E. Static Migration and Error Coverage

The static bank uses an explicit white baseplate and disallows unsupported floating states or passage through the board. Its migration ledger records the five routes in Table S4. All resulting repair witnesses replay under the new executor. Figure S1 shows actual current and target states for three examples.

Of the 267 replacements, 250 source targets were not buildable under the new rules, 13 had board-penetrating initial states, and four were duplicate states. Replacements preserve their source links and substitution reasons. The ledger's 3,025 preserved cases include ten repaired initial scenes; "preserved" therefore does not mean that every initial coordinate was unchanged. The 160 controls have a zero-action repair witness.

A complete geometric replay is a necessary check, but does not establish that every question has sufficient visual evidence or comprehensible wording. The full static RGB package and per-question human readability review are still pending. It would be incorrect to interpret the 3,292/3,292 replay count as a human-validated visual dataset size.

F. Dynamic Mechanisms and Visibility

The mechanism bank contains 68 structures: 42 simplified recognizable objects and 26 smaller structural configurations. Of its 1,872 episodes, 1,588 use complete-object models and 284 use the remaining configurations. There are 160 controls and 1,712 interventions across twelve families (Table S5). This is broader mechanism coverage than the three error families in the community suite, but does not substitute for testing those mechanisms on the larger source-preserving modules.

Machine visibility screens. Pixel differences use a maximum RGB-channel change of at least 24. The reference mechanism audit retains 1,553 interventions as primary machine candidates with at least 2,000 contrast pixels in an available view. Another 136 interventions are classified as action cancellation, and 23 have low contrast; there are no duplicate event states. A visible change in pixel space is not proof that a person can identify the incorrect part or infer a valid correction. Re-rendering and rechecking the realized event is necessary when a different route changes its location or occlusion.

The community reference snapshots use a separate minimum of 128 changed pixels in at least one fixed module view. All 36 reference fault snapshots pass that machine screen. Its lower threshold is an engineering admission criterion, not an experimentally calibrated human perceptual threshold. The two visual admission rules must not be pooled as one validated population.

Splits and difficulty. The mechanism bank groups sources into 43 families for development and evaluation separation. Its balanced development subset has 108 faults (nine per family) and 42 controls; the evaluation subset has 168 faults (fourteen per family) and 72 controls, where "family" in these counts means an error mechanism. Source-family separation applies within this bank, not across every artifact in the repository. The community suite is a public development collection and has no undisclosed held-out source-object evaluation in the current report.

Part count, prerequisite depth, access obstruction, and the amount of reconstruction needed after a fault describe different dimensions of difficulty. Neither source-file size nor the labels "roof" and "car" justify an easy/medium/hard ranking without performance evidence. Figure S2 reports case coverage without making that inference.

Table S2. **Complete community suite.** Every row has four static and four dynamic cases: control, missing piece, wrong color, and extra layer. The threshold is a count of installed normal pieces, not a compulsory reference prefix.

Source set	Object	Remaining module	Source parts	Task parts	Threshold
31009	Cottage	Roof section	270	6	3
31027	Blue racer	Upper cabin	67	11	3
31028	Seaplane	Tail assembly	53	7	3
21004	Guggenheim Museum	Roof section	208	12	4
3179	Repair truck	Cabin top and lights	128	5	4
30181	Orange helicopter	Tail assembly	52	7	3
30311	Police helicopter	Rotor assembly	59	10	4
30026	Panda	Head section	61	6	4
6683	Burger stand	Roof and sign	79	7	3
21006	White House	Chimneys and roof details	556	10	4
21007	Rockefeller Center	Upper crown	240	8	7
21011	Brandenburg Gate	Top crest	362	10	6
Total	12 objects	96 cases	2,135	99	—

Table S3. **Alternative legal routes used in the deterministic study.** Reference-position numbers identify pieces; the actual executor uses randomized handles. All routes reach the same normal target. The threshold states differ for nine objects.

Object	Alternative route in reference-position indices	Different threshold state
Cottage	1, 2, 4, 3, 5, 6	Yes
Racer	1, 6, 3, 4, 5, 2, 7, 8, 9, 10, 11	Yes
Seaplane	1, 6, 3, 4, 5, 2, 7	Yes
Museum	1, 2, 3, 5, 4, 6, 7, 8, 9, 10, 11, 12	Yes
Repair truck	1, 2, 3, 5, 4	Yes
Orange helicopter	1, 3, 2, 4, 5, 6, 7	No
Police helicopter	1, 2, 3, 5, 4, 6, 7, 8, 9, 10	Yes
Panda	1, 2, 3, 5, 4, 6	Yes
Burger stand	2, 1, 3, 4, 5, 6, 7	No
White House	1, 2, 3, 5, 4, 6, 7, 8, 9, 10	Yes
Rockefeller Center	1, 2, 3, 5, 4, 6, 7, 8	No
Brandenburg Gate	1, 2, 3, 4, 5, 7, 6, 8, 9, 10	Yes

Table S4. **White-baseplate migration ledger.** Counts refer to questions, not independent objects.

Migration method	Questions
Direct replay	2,144
Verified dismantle/rebuild	834
Witness using held components	37
Corrected board-penetrating initial scene	10
Replaced incompatible source case	267
Total certified questions	3,292

G. Historical Development Evidence

Earlier API car pilot. A prior interface version records three completed episodes with the provider route named `google/gemini-4-argon` on a six-part car (Table S6). These were development traces with seed 41 and a 40-decision cap. They used a different bank and interface, without the community manuals, and cannot populate a result table for the current suite. One control response was invalidly formatted; no illegal physical action was executed. A normal-run trajectory diverged from the fault runs before the intervention, so these traces are not a paired causal estimate of repair difficulty.

Earlier no-progress runs and HTTP 403 provider failures occurred during interface development. An API access failure is not evidence of a model's assembly ability. There are zero paid model episodes in the new community suite and mechanism-v6 release record.

Earlier human check. One recorded participant answered binary change-detection questions correctly on eleven selected, unprompted images; a twelfth item had already exposed its answer and is excluded. Two further reviewers were reported to agree, but have no independent

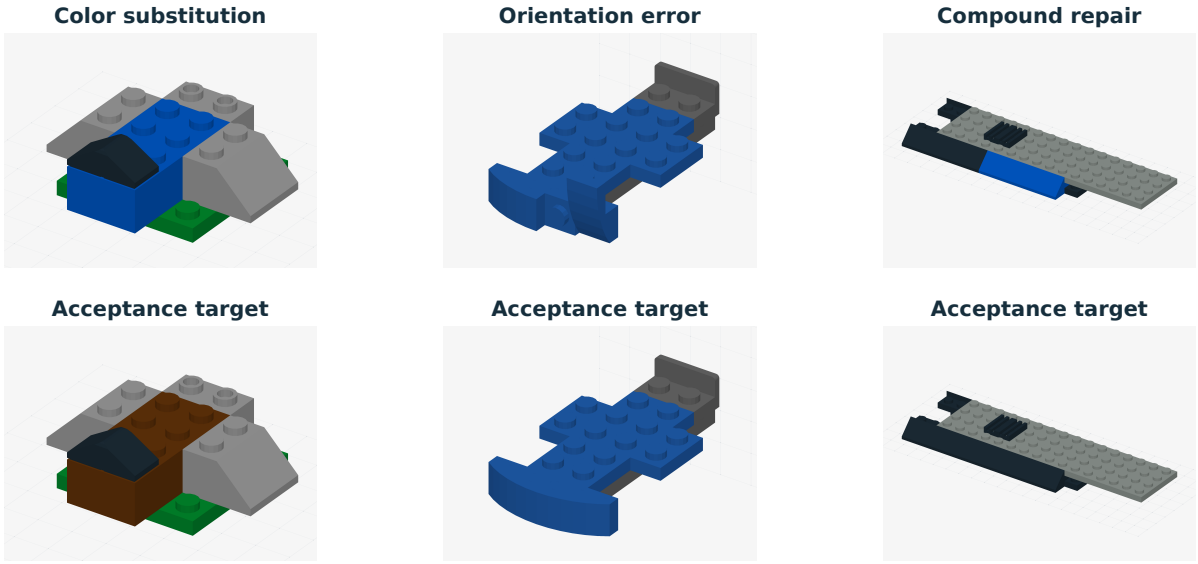


Figure S1. **Static current/target pairs on the white baseplate.** The examples contain a color error (S01609), rotation error (S00254), and compound error (S00001). Each image is rendered from the corresponding certified state at native high resolution. The lower row is the acceptance target; it is not an automatically suggested action.

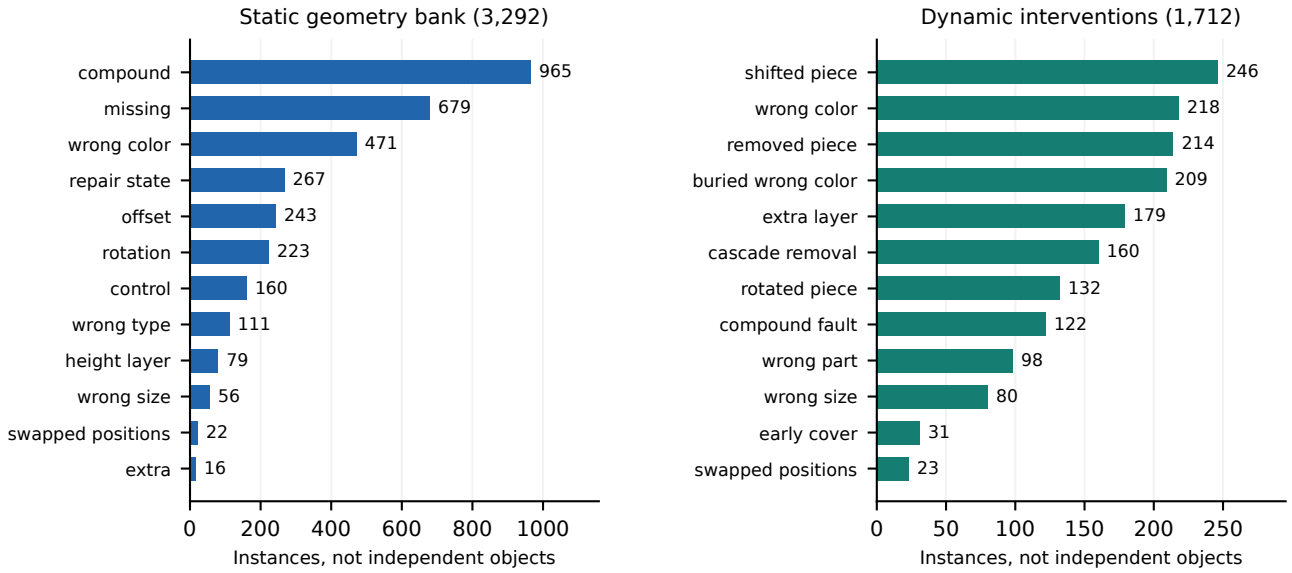


Figure S2. **Coverage of the broader banks.** Static counts include normal controls; dynamic bars show intervention families and exclude 160 controls. Counts describe repeated tasks on shared models, not independent object diversity or measured difficulty.

submission records. This is not a three-rater study, a timed assembly baseline, or validation of all current cases. It supports only a narrow development observation about those eleven images.

Table S5. **Dynamic mechanism counts.** Each episode belongs to one recorded family, including the explicitly compound family.

Condition	Episodes
Normal control	160
Removed piece	214
Shifted piece	246
Cascaded removal	160
Extra layer	179
Early cover	31
Wrong color	218
Wrong size	80
Rotated piece	132
Swapped positions	23
Wrong part type	98
Buried color error	209
Compound fault	122
Total	1,872

Table S6. **Archived pilot only.** Costs are the recorded charges for these individual calls, not current prices or a projected experiment budget. All three completed the older car task.

Condition	Decisions	Legal moves	USD
Control	18	8	0.1950043
Extra layer	14	7	0.2022685
Removed piece	14	7	0.2140114

H. Experiments Still Required for General Claims

The reported executor diagnostics support the stated protocol results. A claim about current agent capabilities requires additional empirical work, listed here to distinguish proposed measurements from collected results.

First, independent participants should complete normal and faulty modules using the same manual, camera access, and one-part interface. Record completion, fault exposure, recovery, action count, time, and ambiguous observations. Validate the realized fault views, not just a paper illustration. Second, compare several model families, including an accessible open-weight baseline where practical, on a source-disjoint test collection. Record exact provider model identifiers, dates, prompts, seeds, retry policy, images, decision caps, and token usage. An interface smoke test should precede the full run; it must not become a post-hoc filter that discards hard cases.

Third, branch common execution prefixes into control and fault trials. Measure coverage, overall completion, conditional recovery, illegal attempts, inspections, latency, and cost. Aggregate by source object and report uncertainty over sources and seeds rather than treating every nearly identical snapshot as independent. Distinguish errors already present before the intervention from failures caused by the edit.

Fourth, ablate temporal observations, manual availability, candidate previews, and legality feedback. A symbolic or expected-state condition can diagnose the visual bottleneck, but must remain clearly privileged. Evaluate unseen module sources and more obstruction/reconstruction mechanisms before making a claim about general human-like assembly. Determine the full budget from measured pilot token use under the actual chosen model prices and observation size; the archived three-car charges are not a valid cost forecast for the new interface.

I. Rendering, Attribution, and Physical Limits

Publication images. All brick scenes in the paper are rendered from original geometry at 3200×2400 pixels, with $2\times$ supersampling in each dimension before downsampling. Composite figure exports use 600 dpi. PDF and SVG retain vector labels, arrows, tables, and process boxes. The paper's larger renders do not replace the 900×660 images received by the current community interface. Orange boxes and manual highlights are presentation annotations and are disclosed in the relevant captions. They are not supplied as automatic fault detections.

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Physical scope. The environment checks rigid geometry, supported connections, and straight manipulation paths. Unsupported surrounding parts are retained as conservative collision bounds rather than deleted, which can reject some motions that a more precise mesh would allow. The preassembled context is fixed. The executor does not model deformation, insertion force, frictional stability, accidental slipping, articulated mechanics, or a robot gripper. Consequently, "legal" and "recoverable" refer to this geometric contract. Neither the protocol study nor its renders establish real-robot performance.